

AI HACKATHON

Agentic Engineering using GitHub Copilot

Today's speakers



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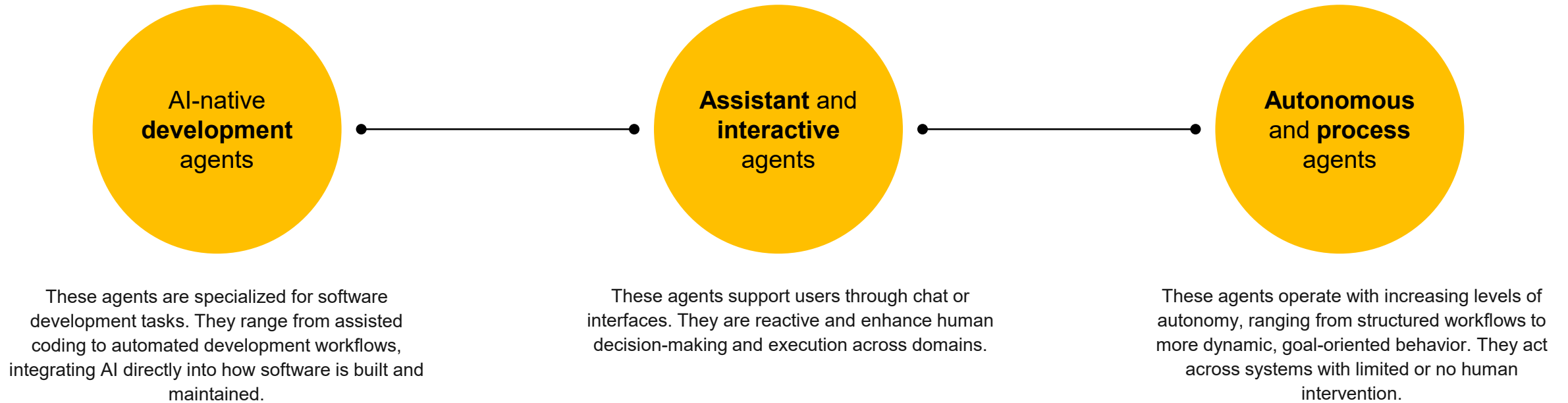
Søren Thestrup

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Data & AI, Denmark

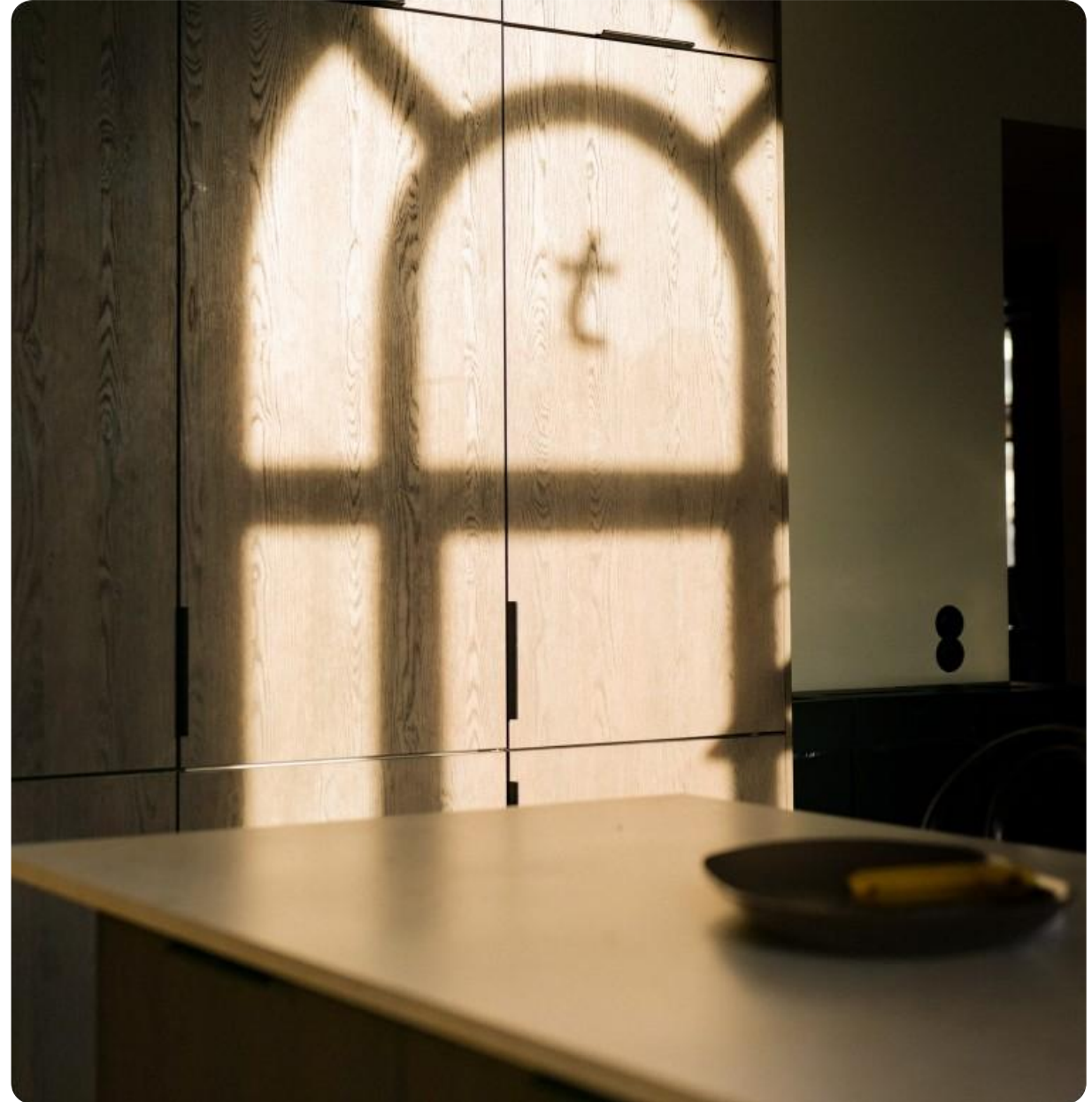
soren.thestrup@twoday.com

The way we deliver and
operate is in change

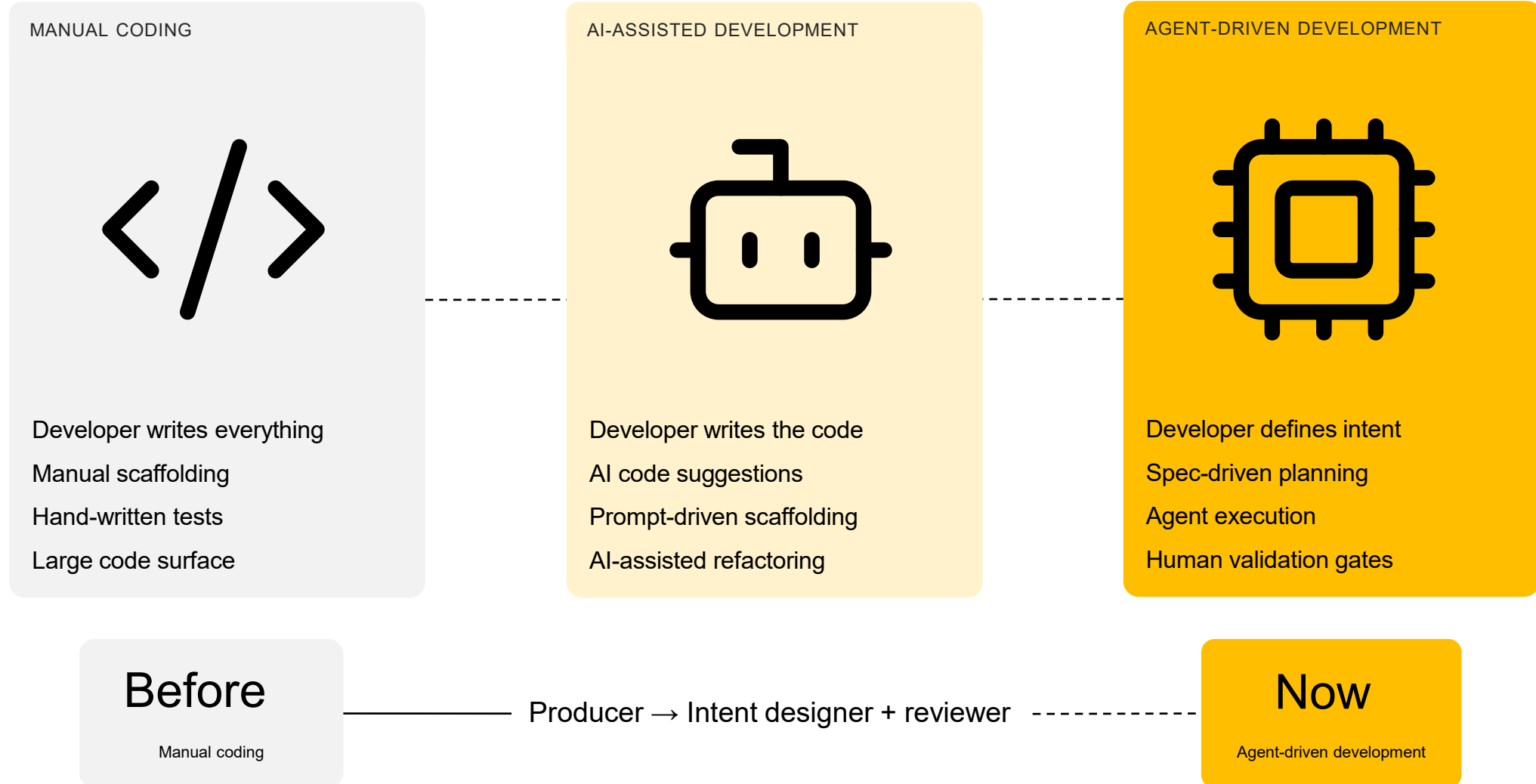
Agentic **design** and **control** in practice



Introduction to Agentic Engineering



How development changes



Different levels of agentic engineering



Prototype, proof-of-concepts or explorative solutions with no focus on security

Build using vibe-coding tools like Lovable, GitHub Spark, Figma Make, etc.

Not recommended for production but great for testing ideas, working on different workflow, A/B testing. Can be done by most people in the organization without any technical knowledge including knowledge about enterprise security, governance, etc.



Pilots, MVPs, or small internal tools with minimum focus on scalability and governance

Build using pro-coding tools combined with agentic tools like GitHub Copilot, Claude Code, etc. and using structured knowledge, workflows, and reusable skills

Ready for production but with limited focus on security, governance, and scalability. Build using structured knowledge, reusable components and skills. Development is handled by agents but validated by developers. Optimal for small to medium greenfield and brownfield projects.



Structured, predictable, scalable, and governed enterprise solutions

Build using pro agentic tools combined using structured knowledge, workflows, reusable skills, and enterprise governance patterns

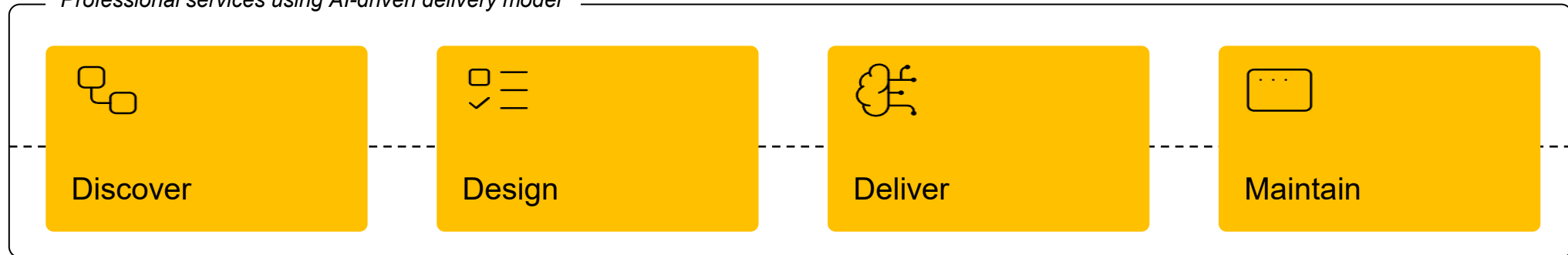
Ready for production with extensive focus on security, governance, and scalability. Build using structured knowledge, workflows, reusable components, and skills. Development is handled by agents but validated by developers. Optimal for medium to large greenfield and brownfield projects with mixed tech stack including modernization.

Delivery of enterprise software solutions using an agent-driven approach

Roles

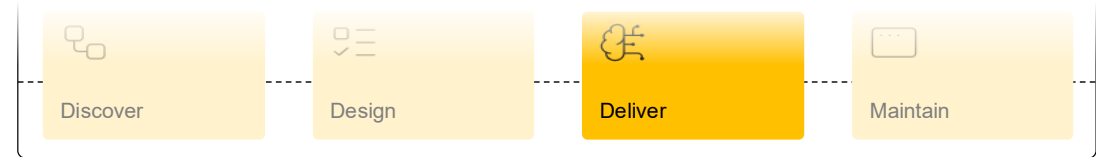
Innovation lead, Discovery facilitator, UX specialist, AI engineer, QA approver, Project orchestrator

Professional services using AI-driven delivery model

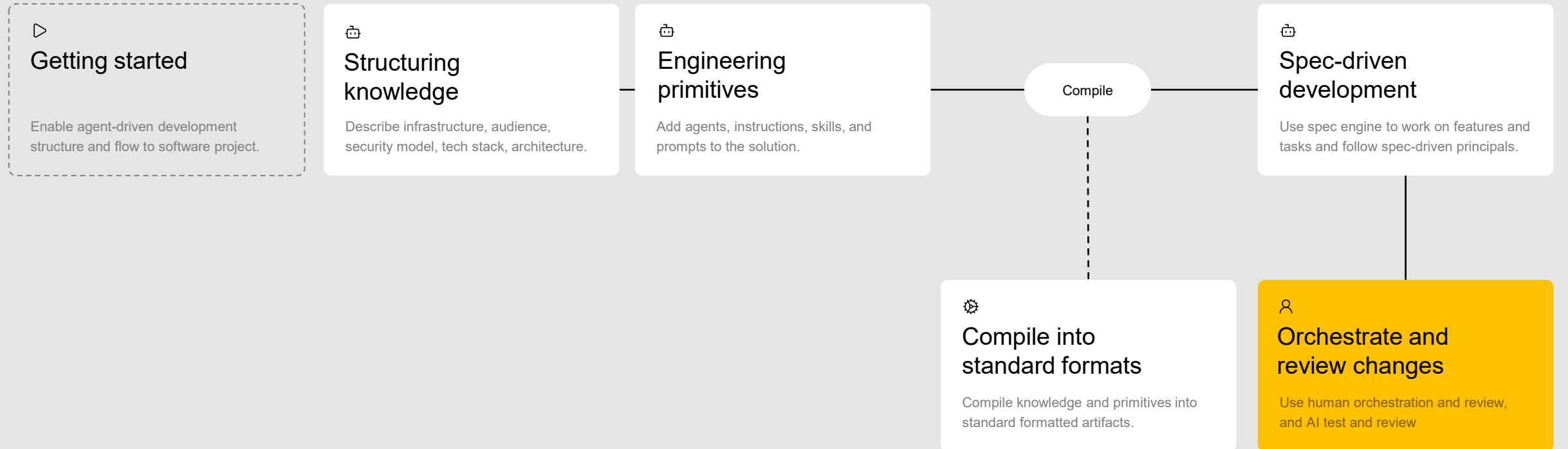


Deliver using agent-driven development

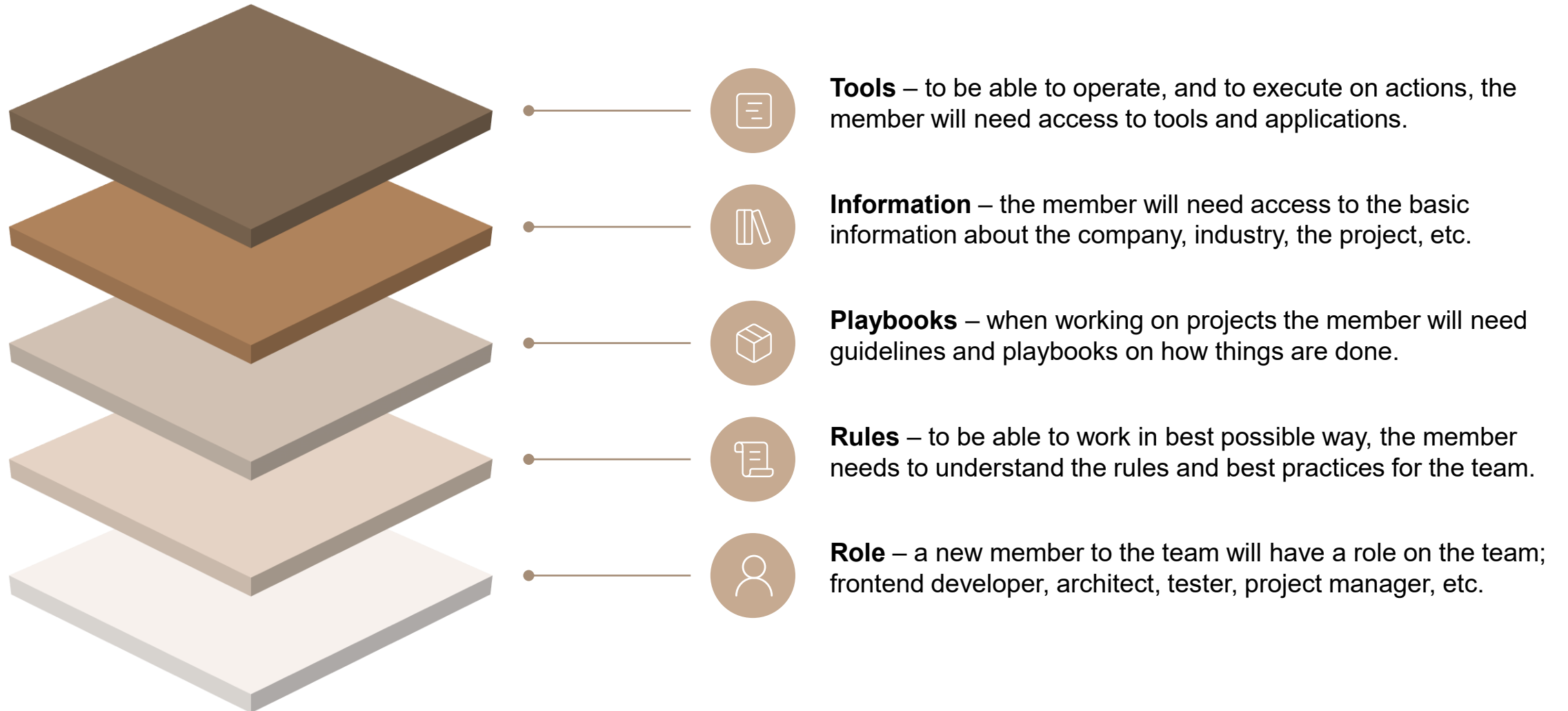
Structured, AI-orchestrated processes where the developer defines intent, specifications guide planning, agents execute tasks, and human validation gates ensure control and quality across the end-to-end flow. Designed to be predictable, reusable, team-scalable, and governed.



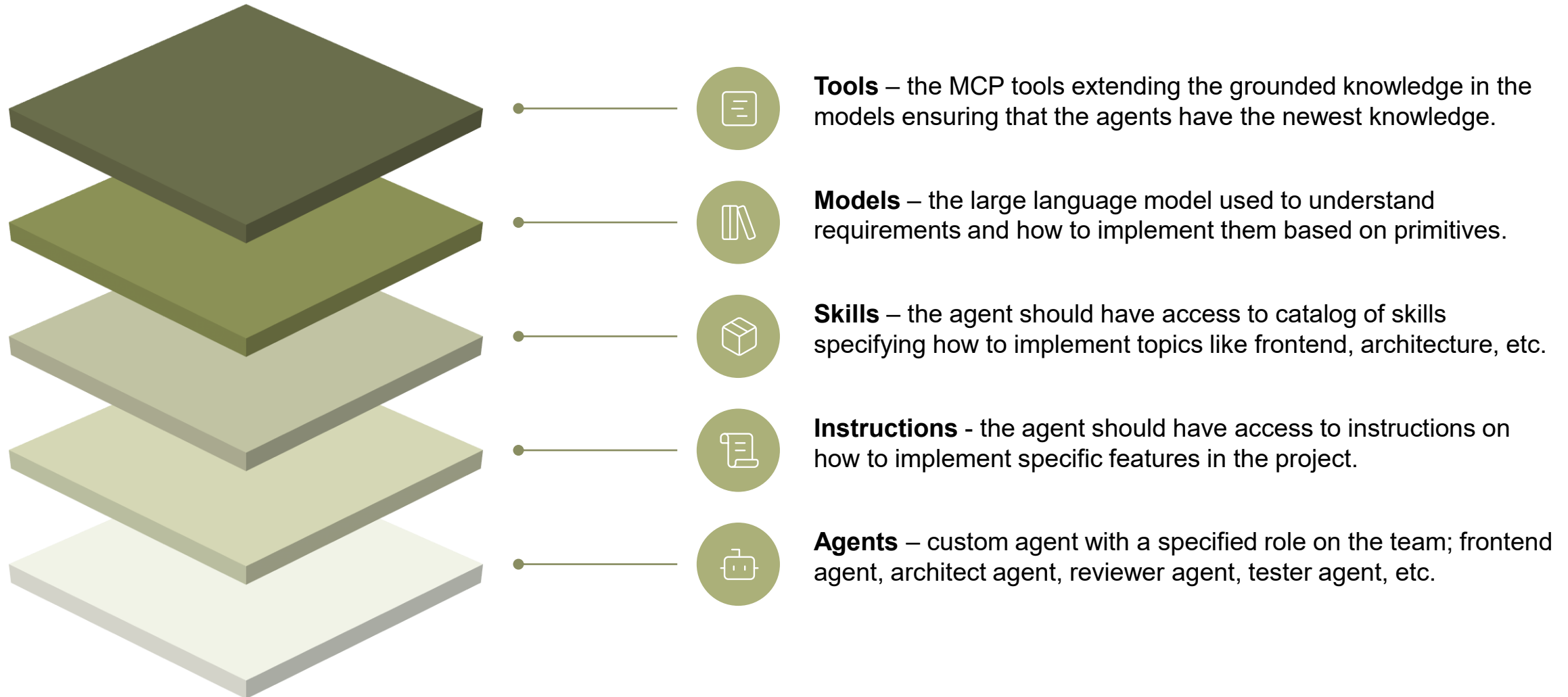
DEVELOPMENT FLOW



Onboarding a new **member** to the team



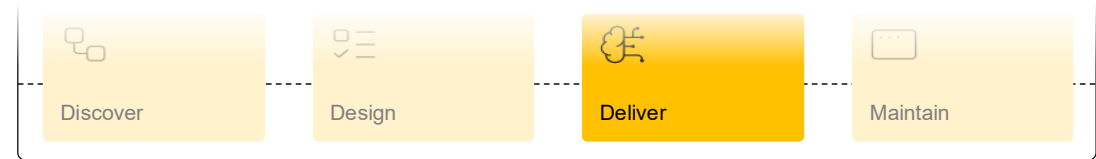
Onboarding a new **agent** to the team



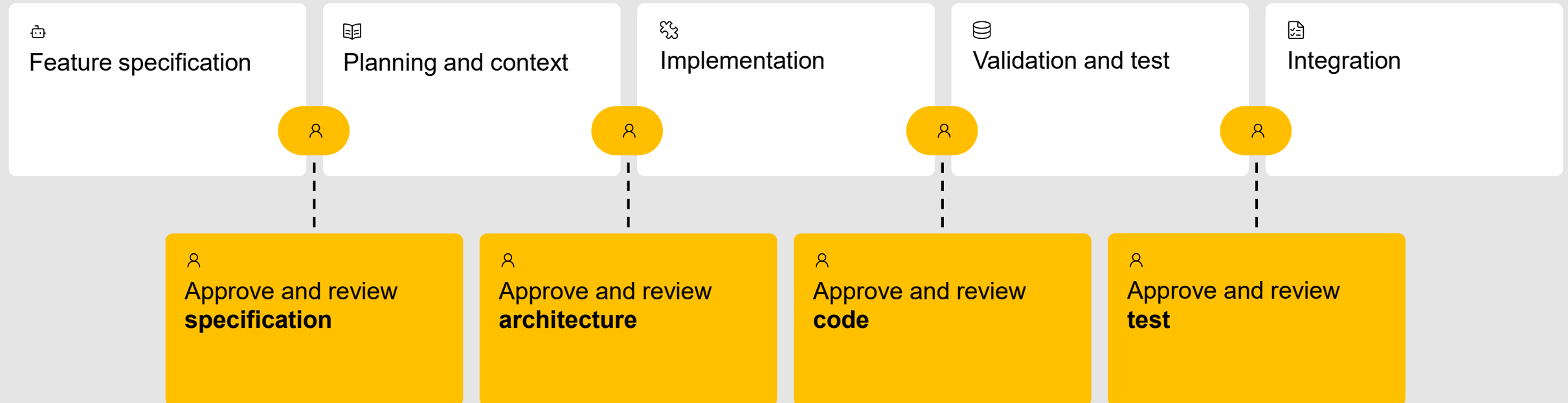
Spec-driven development

Follow spec-driven principals: Change => Implement => Archive.

Use human orchestration and review, and AI test and review



SPEC-DRIVEN DEVELOPEMENT



Agentic building blocks

Primitives are the core configurable elements that developers iteratively refine to ensure reliable outcomes.

AGENTIC BUILDING BLOCKS



Engineering primitives

Add agents, instructions, skills, and prompts to the solution.



Agents

Reusable agents that orchestrate the role, permissions and operating boundaries.



Instructions

Scoped rules and the standards that the agent must follow and in which context.



Skills

Capabilities the agent can use within specific areas and context windows.



Hooks

Capabilities the agent can use within specific areas and context windows.



Orchestration

Use spec engine to work on features and tasks and follow spec-driven principals.



Specifications

Source of truth covering intent, constraints and acceptance criteria.



Memory

Decisions, notes and state maintained and carried forward.



MCP tools

Use spec engine to work on features and tasks and follow spec-driven principals.



Plugins

Decisions, notes and state maintained and carried forward.

Predictable, reusable, and scale across teams

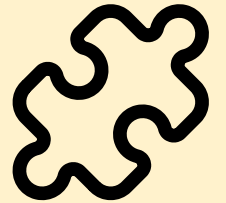
Development time
decrease with
+50%

**01
10**

Quality and
security up with
+20%

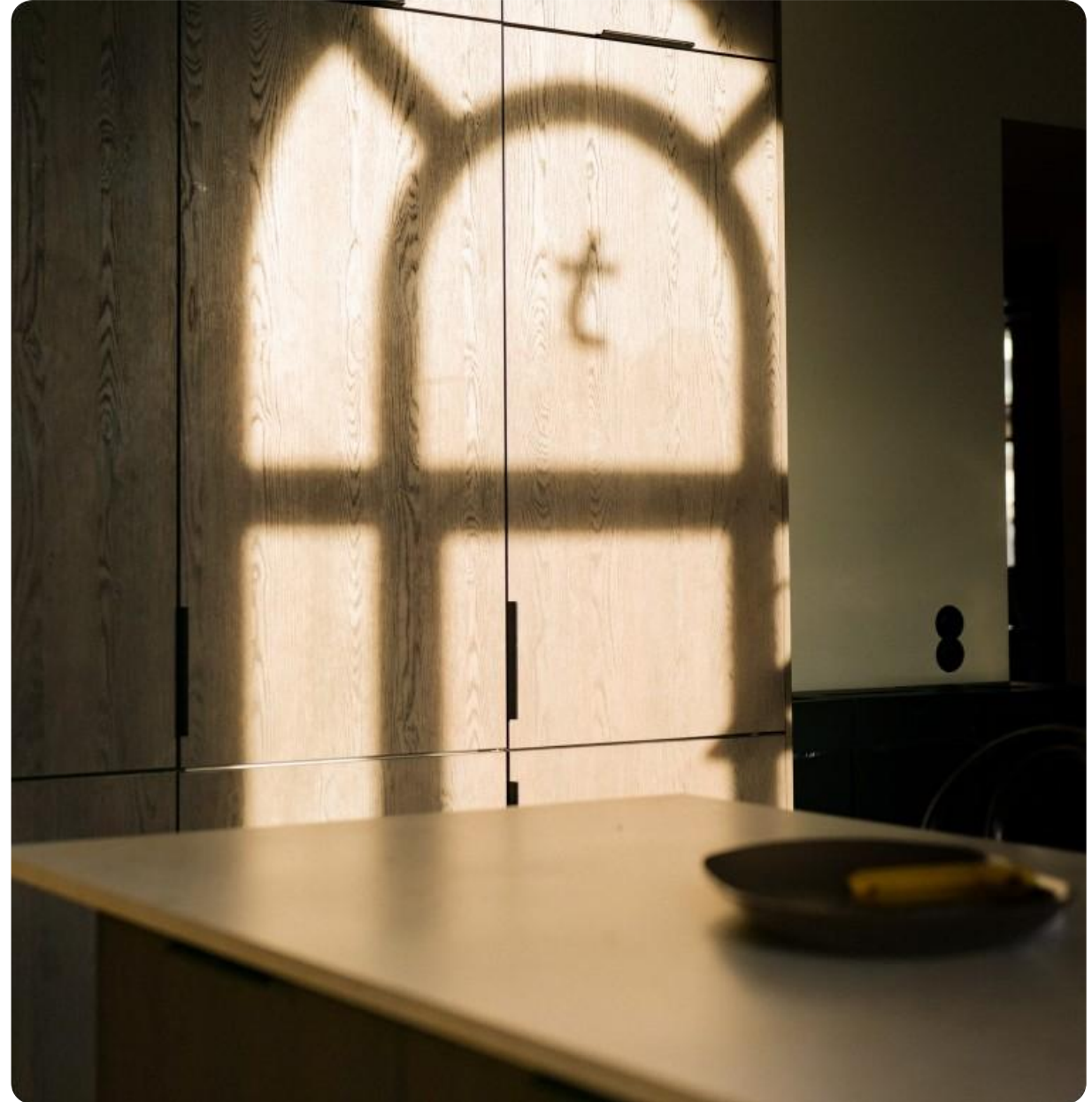


Improve
standardization
>50%



Happy developers
+20%

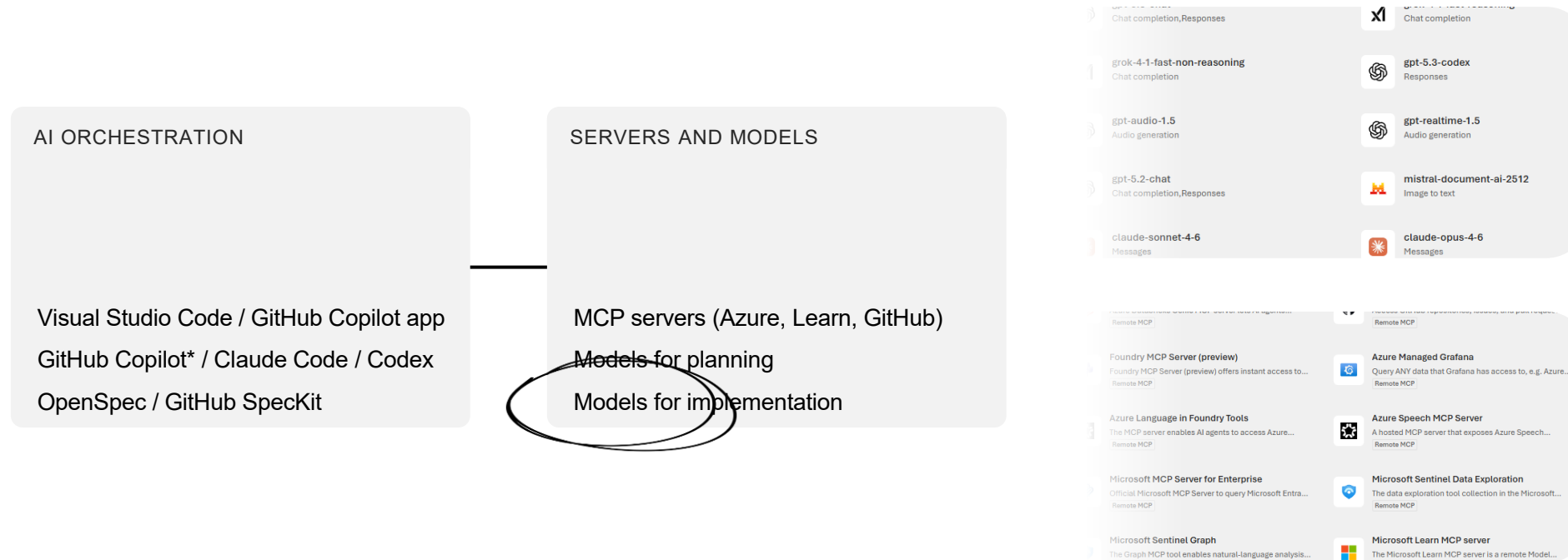
Introduction to GitHub Copilot





GitHub Copilot

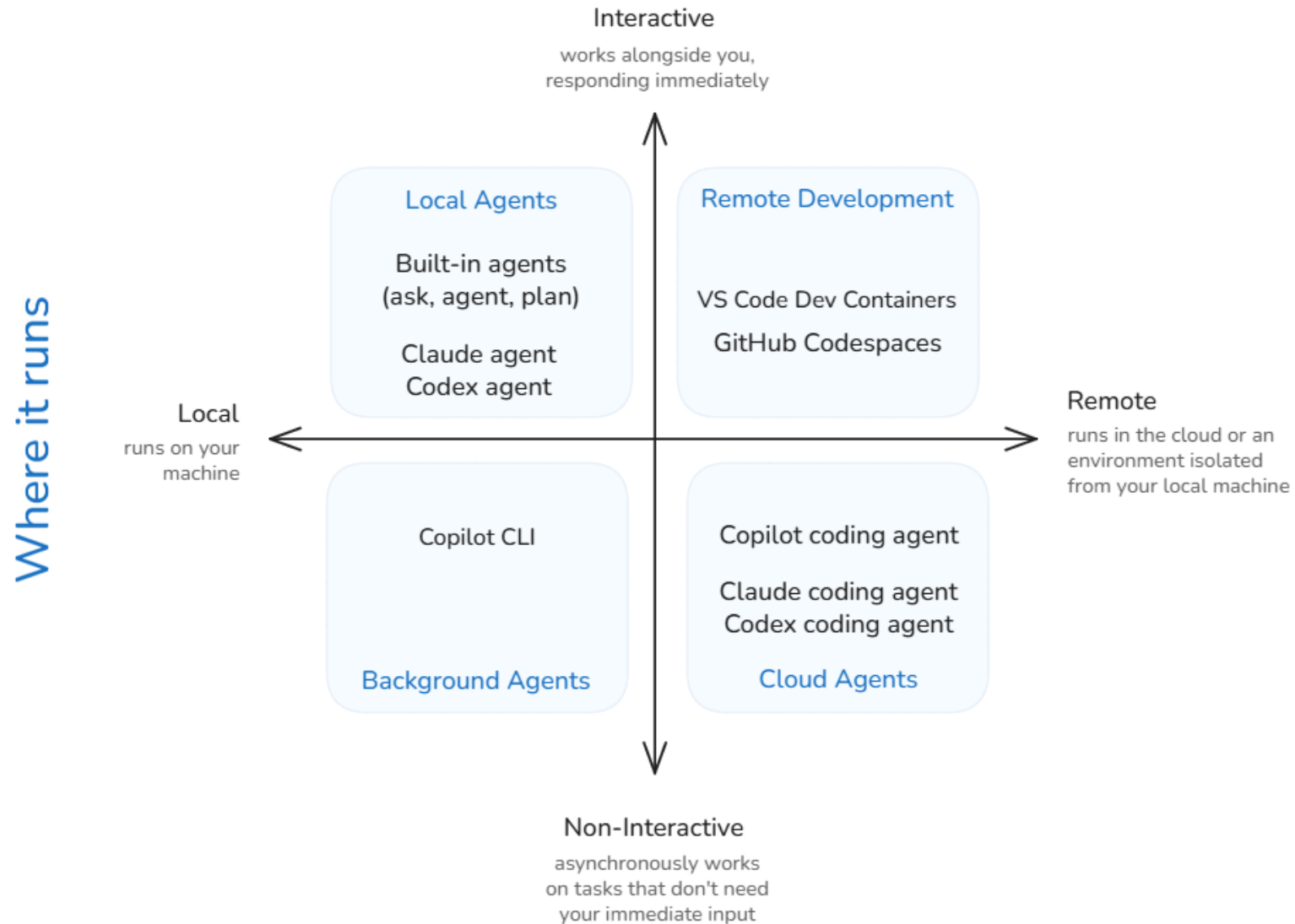
Stay on top of your AI tools and resources



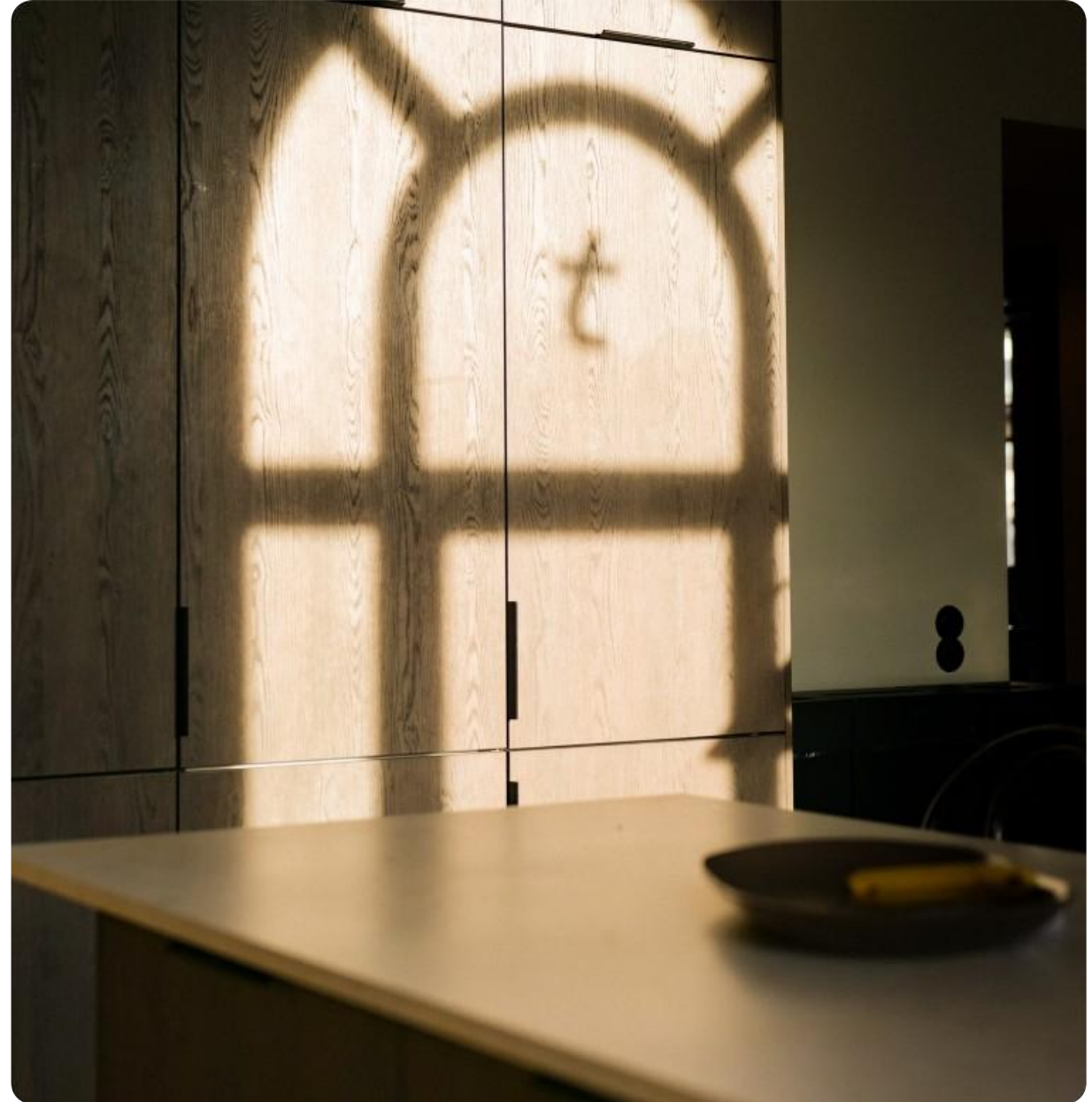
* Business (300 PRPM) or Enterprise (1500 PRPM)

Agent sessions

How you collaborate



Agentic Incident Service case



[https://github.com/thomasmartinsen/
event-hackathon-20260528](https://github.com/thomasmartinsen/event-hackathon-20260528)

Agentic Incident Service – case topics

GitHub

<https://github.com/thomasmartinsen/event-hackathon-20260528>

Agentic Incident Service



Reliable agent-
driven development



Orchestrator
efficiency



Testing as an
evaluation primitive



Security and
resilience

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